Started on	Wednesday, 26 August 2026, 6:57 PM
State	Finished
Completed on	Wednesday, 26 August 2026, 7:20 PM
Time taken	22 mins 17 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

Reset answer

```c
#include<stdio.h>
int count =0;
int factor (int num){

    for(int i=1;i<=num;++i){
        count+=1;

        if (num%i==0)
            count+=1;   //num%i==0 (comparison)
        count++;    // i  updation

    }count++;   //false
    return count;
}

int main(){
    int n;
    scanf ("%d",&n);

    printf("%d",factor(n));
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔

	Input	Expected	Got	
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.